

3.2 Matrix Multiplication: The Second Perspective



Core skills you'll master in this section:

- ▶ The outer-product-expansion perspective: matrix multiplication can be expressed as a sum of several outer-product matrices.
- ▶ Each outer product is a rank-one matrix.
- ▶ A rank-one matrix has row and column multiple-relationships: this reflects linear dependence.
- ▶ The outer product of a zero vector results in a zero matrix.

The second perspective on matrix multiplication is a decomposition that writes the left-hand matrix as a set of column vectors and the right-hand matrix as a set of row vectors. This perspective reveals another essential nature of matrix multiplication — a sum of outer products between column vectors and row vectors. Through this decomposition, we can gain a deeper understanding of the structure and meaning of matrix multiplication.

Superposition

For the matrix multiplication $\mathbf{A} @ \mathbf{B}$, as shown in Figure 1, write the left-hand matrix $\mathbf{A}$ as a set of column vectors

$$\mathbf{A}_{m \times p} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_p]$$

Among these, column vector $\mathbf{a}_k$ ($k = 1, 2, \dots, p$) has shape $m \times 1$.

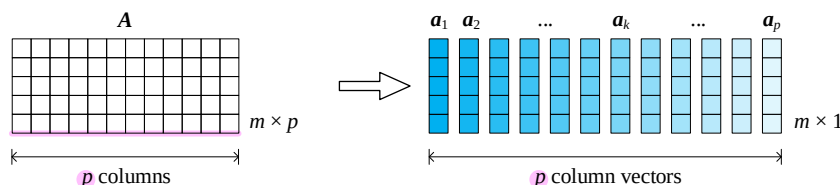


Figure 1. Write the Left-Hand Matrix $\mathbf{A}$ as a Set of Column Vectors

As shown in Figure 2, write the right-hand matrix $\mathbf{B}$ as a set of row vectors

$$\mathbf{B}_{p \times n} = \begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \vdots \\ \mathbf{b}^{(p)} \end{bmatrix}$$

Among these, row vector $\mathbf{b}^{(k)}$ has shape $1 \times n$.

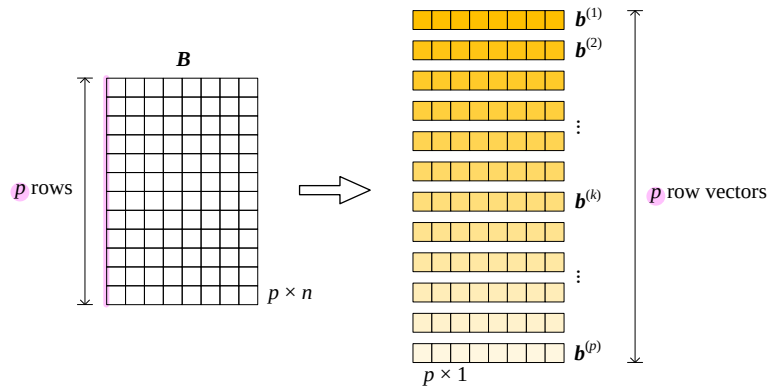


Figure 2. Write the Right-Hand Matrix $\mathbf{B}$ as a Set of Row Vectors

In this way, the matrix multiplication $\mathbf{C} = \mathbf{A} @ \mathbf{B}$ can be written as

$$\mathbf{A}_{m \times p} @ \mathbf{B}_{p \times n} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p \end{bmatrix} @ \begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \vdots \\ \mathbf{b}^{(p)} \end{bmatrix} = \mathbf{a}_1 @ \mathbf{b}^{(1)} + \mathbf{a}_2 @ \mathbf{b}^{(2)} + \cdots + \mathbf{a}_p @ \mathbf{b}^{(p)} = \sum_{k=1}^p \mathbf{a}_k @ \mathbf{b}^{(k)}$$

Did you notice something surprising — we've written the matrix multiplication as a sum of matrices that all share the same shape! This is shown specifically in Figure 3.

That is, the matrix multiplication $\mathbf{A} @ \mathbf{B}$ is transformed into a superposition of p matrices. Here p corresponds to the number of columns of $\mathbf{A}$ and the number of rows of $\mathbf{B}$ — p is the dimension that gets “canceled out.”

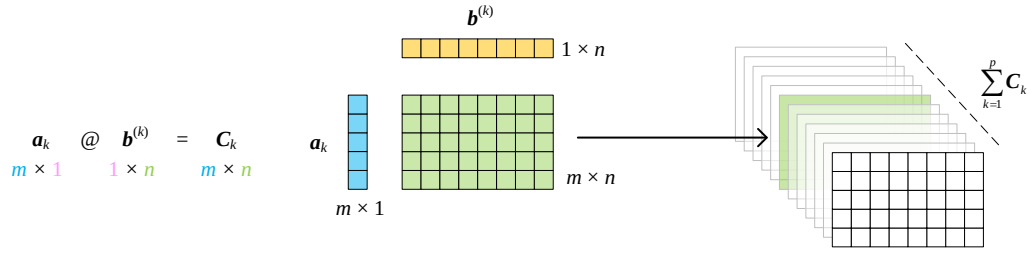
Let

$$\mathbf{C}_k = \mathbf{a}_k @ \mathbf{b}^{(k)}$$

$\mathbf{C}_k$ is exactly the outer product mentioned in the previous section. In addition, notice that matrix $\mathbf{C}$ and $\mathbf{C}_k$ have exactly the same shape — both are $m \times n$.

Looking at the matrix shapes, $(m \times 1) @ (1 \times n)$ — the (1) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape $(m \times n)$.

This, then, is the second perspective on the rule of matrix multiplication — outer product expansion (outer product expansion).

Figure 3. $A @ B$ Can Be Written as a Sum of Matrices with the Same Shape

The First Example

Let's return to the matrix-multiplication example from the first section of this chapter.

$$A @ B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

The left-hand matrix A is written as column vectors, and the right-hand matrix B is written as row vectors; then $A @ B$ can be written as

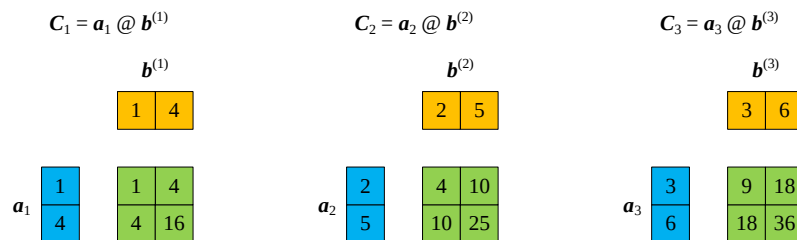
$$A @ B = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} b^{(1)} \\ b^{(2)} \\ b^{(3)} \end{bmatrix} = a_1 @ b^{(1)} + a_2 @ b^{(2)} + a_3 @ b^{(3)}$$

Substituting specific values

$$\begin{aligned} a_1 @ b^{(1)} &= \begin{bmatrix} 1 \\ 4 \end{bmatrix} @ [1, 4] = \begin{bmatrix} 1 \times 1 & 1 \times 4 \\ 4 \times 1 & 4 \times 4 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 4 & 16 \end{bmatrix} \\ a_2 @ b^{(2)} &= \begin{bmatrix} 2 \\ 5 \end{bmatrix} @ [2, 5] = \begin{bmatrix} 2 \times 2 & 2 \times 5 \\ 5 \times 2 & 5 \times 5 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 25 \end{bmatrix} \\ a_3 @ b^{(3)} &= \begin{bmatrix} 3 \\ 6 \end{bmatrix} @ [3, 6] = \begin{bmatrix} 3 \times 3 & 3 \times 6 \\ 6 \times 3 & 6 \times 6 \end{bmatrix} = \begin{bmatrix} 9 & 18 \\ 18 & 36 \end{bmatrix} \end{aligned}$$

Figure 4 shows how to compute C_1 , C_2 , and C_3 respectively.

Looking at the matrix shapes, $(2 \times 1) @ (1 \times 2)$ — the (1) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (2×2) .

Figure 4. Computing C_1 , C_2 , and C_3

As shown in Figure 5, the matrix multiplication $C = A @ B$ can be written as a superposition of three matrices

$$C = \begin{bmatrix} 1 & 4 \\ 4 & 16 \end{bmatrix} + \begin{bmatrix} 4 & 10 \\ 10 & 25 \end{bmatrix} + \begin{bmatrix} 9 & 18 \\ 18 & 36 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

$$C_1 = a_1 @ b^{(1)} \quad C_2 = a_2 @ b^{(2)} \quad C_3 = a_3 @ b^{(3)} \quad = \quad C$$

Figure 5. C_1, C_2, C_3 – Superposition of Three Matrices

The Second Example

The left-hand matrix B is written as column vectors, and the right-hand matrix A is written as row vectors; then $B @ A$ can be written as

$$B @ A = \begin{bmatrix} b_1 & b_2 \end{bmatrix} \begin{bmatrix} a^{(1)} \\ a^{(2)} \end{bmatrix} = b_1 @ a^{(1)} + b_2 @ a^{(2)}$$

Substituting specific values

$$b_1 @ a^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} @ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 \times 1 & 1 \times 2 & 1 \times 3 \\ 2 \times 1 & 2 \times 2 & 2 \times 3 \\ 3 \times 1 & 3 \times 2 & 3 \times 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$b_2 @ a^{(2)} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} @ \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 4 \times 4 & 4 \times 5 & 4 \times 6 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 6 \times 4 & 6 \times 5 & 6 \times 6 \end{bmatrix} = \begin{bmatrix} 16 & 20 & 24 \\ 20 & 25 & 30 \\ 24 & 30 & 36 \end{bmatrix}$$

Figure 6 shows how to compute D_1 and D_2 .

Looking at the matrix shapes, $(3 \times 1) @ (1 \times 3)$ – the (1) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (3×3) .

$$D_1 = b_1 @ a^{(1)} \quad D_2 = b_2 @ a^{(2)}$$

Figure 6. Computing D_1 and D_2

As shown in Figure 7, the matrix multiplication $D = B @ A$ can be written as a superposition of two matrices

$$\mathbf{D} = \mathbf{b}_1 @ \mathbf{a}^{(1)} + \mathbf{b}_2 @ \mathbf{a}^{(2)} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} + \begin{bmatrix} 16 & 20 & 24 \\ 20 & 25 & 30 \\ 24 & 30 & 36 \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

$$\mathbf{D}_1 = \mathbf{b}_1 @ \mathbf{a}^{(1)} \quad \mathbf{D}_2 = \mathbf{b}_2 @ \mathbf{a}^{(2)}$$

1	2	3
2	4	6
3	6	9

$$+$$

16	20	24
20	25	30
24	30	36

$$=$$

17	22	27
22	29	36
27	36	45

Figure 7. $\mathbf{D}_1$ and $\mathbf{D}_2$ — Superposition of Two Matrices

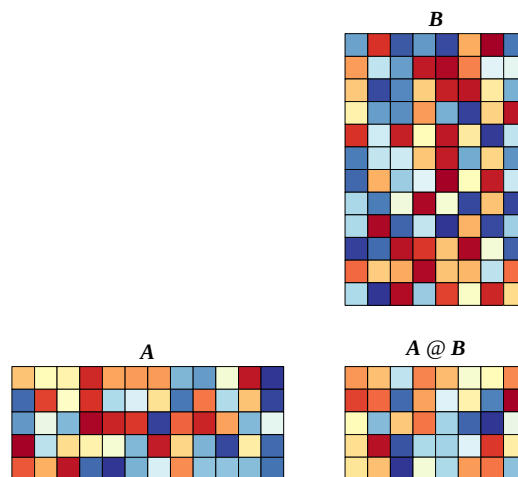
LA_03_02_01.ipynb completes the two matrix-multiplication operations above using the second perspective on matrix multiplication — please study it on your own.

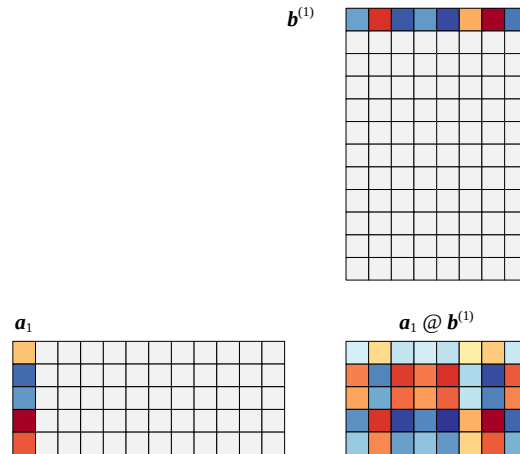
Visualizing the Second Perspective on Matrix Multiplication with a Heatmap

Next, let's use a heatmap to visualize the second perspective on matrix multiplication.

Given the matrix multiplication shown in Figure 8.

Figure 9 uses a heatmap to show how to compute $\mathbf{a}_1 @ \mathbf{b}^{(1)}$.

Figure 8. Matrix Multiplication $\mathbf{A} @ \mathbf{B}$ Heatmap, First Perspective

Figure 9. Heatmap Showing $\mathbf{a}_1 @ \mathbf{b}^{(1)}$

Since $\mathbf{A}$ has 12 columns ($\mathbf{B}$ has 12 rows), there are matrix multiplications like $\mathbf{a}_1 @ \mathbf{b}^{(1)}$, 12 in total, as shown in Figure 10.

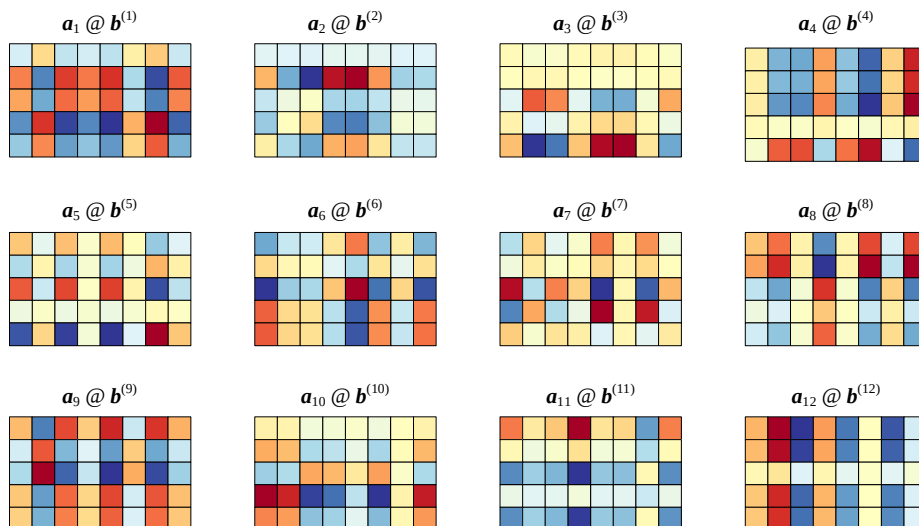
Figure 10. 12 $\mathbf{a}_k @ \mathbf{b}^{(k)}$

Figure 10, the result of adding these 12 matrices together is $\mathbf{A} @ \mathbf{B}$.



LA_03_02_02.ipynb draws the heatmap above — please study it on your own.

Multiple Relationships

As shown in Figure 11, given $\mathbf{a}$ and $\mathbf{b}$ — two column vectors — the matrix multiplication $\mathbf{a} @ \mathbf{b}^T$ has row and column multiple-relationships

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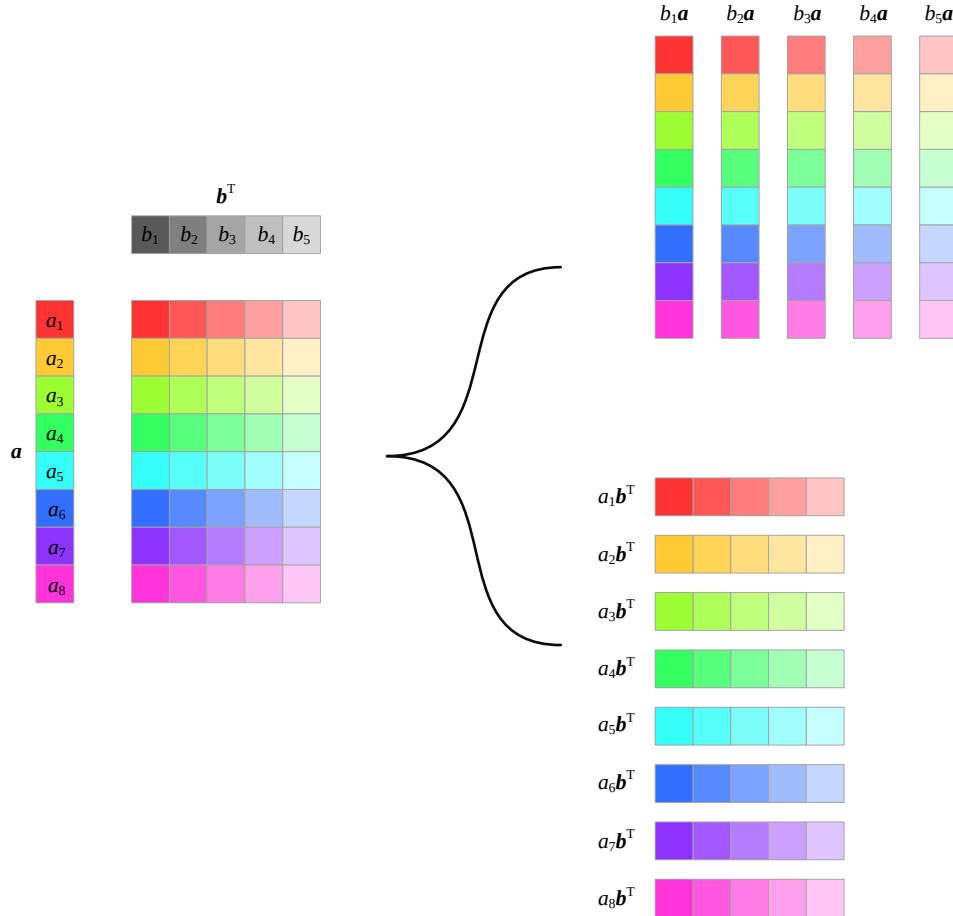
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$$\mathbf{a} @ \mathbf{b}^T = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} @ \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_m b_1 & a_m b_2 & \cdots & a_m b_n \end{bmatrix}$$

Figure 11. Matrix Multiplication $\mathbf{a} @ \mathbf{b}^T$ Has Row and Column Multiple-Relationships

Let's first look at the columns of $\mathbf{a} @ \mathbf{b}^T$, that is

$$\mathbf{a} @ \mathbf{b}^T = \mathbf{a} @ \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} b_1 \mathbf{a} & b_2 \mathbf{a} & \cdots & b_n \mathbf{a} \end{bmatrix}$$

Expanding, we get

$$\mathbf{a} @ \mathbf{b}^T = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} b_1 \times \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} & b_2 \times \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} & \cdots & b_n \times \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \end{bmatrix}$$

Let's now look at the rows of $\mathbf{a} @ \mathbf{b}^T$

$$\mathbf{a} @ \mathbf{b}^T = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \mathbf{b}^T = \begin{bmatrix} a_1 \mathbf{b}^T \\ a_2 \mathbf{b}^T \\ \vdots \\ a_m \mathbf{b}^T \end{bmatrix}$$

Expanding, we get

$$\mathbf{a} @ \mathbf{b}^T = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} a_1 \times [b_1 & b_2 & \cdots & b_n] \\ a_2 \times [b_1 & b_2 & \cdots & b_n] \\ \vdots \\ a_m \times [b_1 & b_2 & \cdots & b_n] \end{bmatrix}$$

Precisely because of this row/column multiple-relationship, we call this kind of outer product $\mathbf{a} @ \mathbf{b}^T$ a rank-one matrix (rank-one matrix).

Every row of a rank-one matrix is a multiple of the same row vector; every column of a rank-one matrix is a multiple of the same column vector.

This shows that the rows and columns of a rank-one matrix are linearly dependent on each other — that is, the rank (rank) of the matrix is 1. Simply put, the rank of a matrix is the maximum number of linearly independent vectors among its rows or columns, equivalent to the maximum number of nonzero, linearly independent row or column vectors in the matrix. This is a topic this book will explore in depth later.

 Note that if $\mathbf{a}$ or $\mathbf{b}$ is the zero vector $\mathbf{0}$, then $\mathbf{a} @ \mathbf{b}^T$ results in the zero matrix $\mathbf{0}$, in that case, the matrix $\mathbf{a} @ \mathbf{b}^T$ has rank 0.

Returning to the earlier example, you can already see the multiple-relationship between the column vectors

$$\mathbf{b}_1 @ \mathbf{a}^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} @ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} & 2 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} & 3 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \end{bmatrix}$$

Now look at the multiple-relationship between the row vectors

$$\mathbf{b}_1 @ \mathbf{a}^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} @ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \\ 2 \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \\ 3 \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \end{bmatrix}$$

A Custom Python Function to Compute Matrix Multiplication: The Outer-Product Perspective

Code 1 uses a custom function to complete the matrix multiplication using the outer-product perspective. Let's go over the key statements below.

 Defines a function named `matrix_multiplication_outer`, meaning “compute matrix multiplication using the outer-product approach”; a function packages a block of code into a reusable tool, and this one takes matrix A and matrix B as inputs.

b Starts a loop, from $k = 0$ up through p – that is, we take out every column of A one at a time, and correspondingly take out every row of B ; this sets up the process of computing the “outer product” one column (and row) at a time.

$a_k = A[:, [k]]$ extracts the k -th column of matrix A , turning it into a vertical column vector; $b_k = B[[k], :]$ extracts the k -th row of matrix B . $C += (a_k @ b_k)$ computes the outer product, adding this matrix to the result matrix C – accumulating this step's contribution.

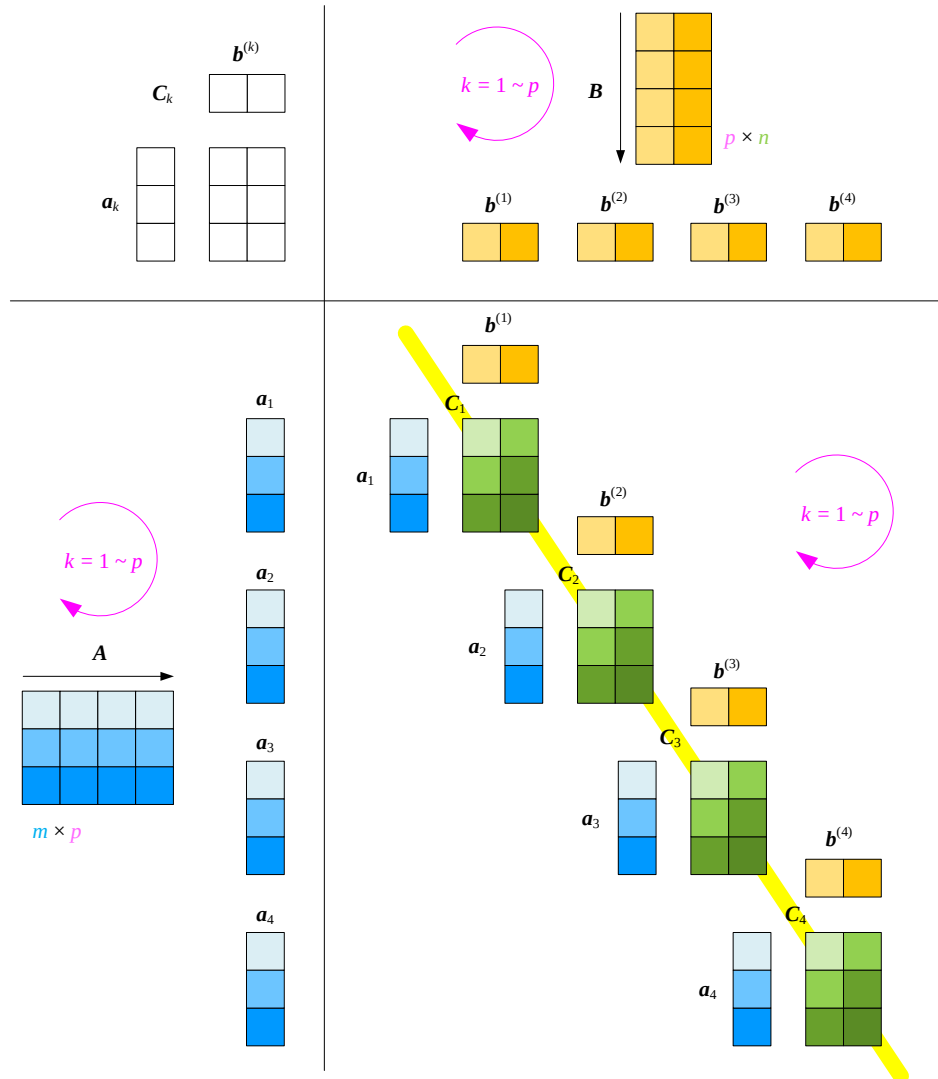


Figure 12. The Rule for the Matrix Multiplication AB , the Outer-Product Perspective

Code 1. A Custom Python Function to Compute Matrix Multiplication, the Outer-Product Perspective | LA_03_02_03.ipynb

```

## 初始化
import numpy as np

## 自定义函数，矩阵乘法第二视角
def matrix_multiplication_outer(A, B):

    # 获取矩阵 A 和 B 的形状
    m, p_A = A.shape
    p_B, n = B.shape

    # 检测矩阵形状是否符合矩阵乘法规则
    if p_A != p_B:
        raise ValueError('Dimensions do not match')

    # 初始化结果矩阵 C，形状 (m, n)，初始值设为 0
    C = np.zeros((m, n))

    for k in range(p_A):
        a_k = A[:, [k]]      # A 的第 k 列，二维数组
        b_k = B[[k], :]      # B 的第 k 行，二维数组
        C += (a_k @ b_k)      # 每次计算外积并加到结果中

    return C

## 矩阵乘法
A = np.array([[1, 2, 3],
              [4, 5, 6]])
B = A.T

## 矩阵乘法
matrix_multiplication_outer(A, B)
matrix_multiplication_outer(B, A)

```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Given the following pair of matrices A and B , use the second-perspective method to compute $A @ B$ and $B @ A$

▶ $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

▶ $A = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

▶ $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad \mathbf{B} = \mathbf{A}^T$$

Q2. Given the following pairs of column vectors, write the matrix multiplication $\mathbf{a} @ \mathbf{b}^T$ as a row/column multiple-relationship.

► $\mathbf{a} = [1, 2]^T, \mathbf{b} = [1, 2]^T$

► $\mathbf{a} = [1, 2]^T, \mathbf{b} = [1, 2, 3]^T$

► $\mathbf{a} = [1, 2, 3]^T, \mathbf{b} = [1, 2]^T$

Q3. What is the rank of a matrix, and is it related to the matrix's size?

Q4. What is linear dependence? How can it be understood geometrically?

Q5. What is linear independence? How can it be understood geometrically?